

MSc Project Plan — Proposal

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Title: Explainable AI for Financial Customer Service (CLARIX)

Problem Outline

Financial institutions in the United Kingdom receive hundreds of thousands of consumer complaints annually. The Financial Ombudsman Service (FOS) handled over 200,000 new cases in 2023–2024 spanning banking, credit, insurance, and debt collection (FOS, 2024). Despite this scale, complaint intake and triage remains predominantly manual in most organisations, producing inconsistent categorisation, delayed escalation of urgent fraud cases, and growing compliance risk under the FCA Consumer Duty (PS22/9, 2023), which legally obliges firms to evidence fair, timely, and consistent complaint outcomes for all retail customers. A second critical gap is explainability: GDPR Article 22 grants individuals the right to explanation when automated decisions affect them (Gunning & Aha, 2019), yet existing platforms such as Salesforce Service Cloud provide complaint classification without regulatory-grade transparency. This project addresses both gaps by building a domain-specific, explainable AI financial customer service platform that automates complaint classification, priority prediction, and sentiment analysis while generating SHAP-based explanations for every AI decision.

Background

The FCA Consumer Duty (PS22/9) came into force in July 2023, requiring firms to evidence that complaint handling delivers measurable good outcomes with explicit scrutiny on consistency and timeliness (FCA, 2023). The EU AI Act (2024) classifies AI systems used in financial service access as high-risk, mandating transparency, human oversight, and auditability. These regulatory developments create a timely and academically grounded research context for an explainable AI complaint triage system in banking and insurance.

Research in Natural Language Processing has established that transformer-based models fine-tuned on financial domain text significantly outperform general-purpose models (Araci, 2019; Devlin et al., 2019). Lundberg and Lee (2017) introduced SHAP, providing model-agnostic feature attribution that satisfies GDPR Article 22 transparency requirements. The application of explainable NLP to financial consumer complaint triage remains underexplored, representing a clear research contribution.

Aims/Objectives

The overarching aim is to design, implement, and evaluate a domain-specific explainable AI financial customer service platform demonstrating that NLP-driven complaint triage can improve classification accuracy and regulatory transparency compared to manual categorisation. The following SMART objectives structure the work:

- O1 — Train and evaluate NLP classification models for complaint category (6-class), priority (3-class), and sentiment (3-class) on the CFPB dataset and FinancialPhraseBank, achieving a minimum weighted F1-score of 0.70 on a held-out test set.
- O2 — Implement SHAP-based explainability for all AI predictions and integrate the full NLP pipeline into a FastAPI backend with core services (authentication, case management, AI inference, chatbot, automation, analytics) verified through integration tests.
- O3 — Develop a server-rendered Next.js web application with role-differentiated portals (Customer, Agent, Supervisor, Manager, Admin), an embedded complaint chatbot, and an interactive analytics dashboard.
- O4 — Produce a comprehensive test suite (unit, integration, and security tests) and package the complete system for local Docker Compose execution with a README and 10-minute demonstration video.

Research Questions

RQ1 (Primary): To what extent can a microservices-based explainable NLP pipeline — combining multi-label financial complaint classification, priority prediction, and SHAP-based explanations — improve the accuracy and regulatory transparency of complaint triage in banking and insurance compared to manual categorisation?

RQ2: How well do the trained NLP classification models generalise to unseen financial complaint text, and what are the main sources of misclassification across the six complaint categories when evaluated on the held-out CFPB test set?

RQ3: What fairness and bias implications arise when applying supervised NLP classification to financial complaint data, and how can SHAP feature attribution surface and support mitigation of linguistic bias in automated triage decisions?

RQ4: How can SHAP-based feature attribution support regulatory audit requirements for automated financial complaint triage under FCA Consumer Duty and GDPR Article 22, and to what extent do the generated explanations satisfy the right to explanation in practice?

Artefact Deliverables

- Modular FastAPI backend with core services orchestrated via Docker Compose with health checks, named volumes, and Nginx reverse proxy.
- Three trained NLP models: complaint category (6-class, CFPB dataset), priority (3-class, engineered labels), sentiment (3-class, FinancialPhraseBank), with SHAP explainability and serialised artefacts served through the AI inference service.
- Server-rendered Next.js (TypeScript) web application with five role-differentiated portals. Agent dashboard surfaces AI predictions and SHAP explanations inline. Embedded complaint chatbot widget.
- Interactive analytics dashboard built into the web application: complaint volume by category, sentiment trend, FCA response compliance rate, and agent performance — powered by PostgreSQL reporting tables and rendered using Recharts.
- Hybrid chatbot: rule-based flow + NLP intent detection across six financial complaint categories, escalation to ticket creation, transcript stored in MongoDB.
- Five domain-specific automation rules delivered via SMTP (MailHog in development).
- Full local Docker Compose orchestration with health checks, named volumes, Nginx reverse proxy, and a detailed README with run instructions.
- Comprehensive test suite: unit, integration, security, performance, ML evaluation evidence.

Artefact Development

Methods for analysis, design and implementation:

An Agile-inspired iterative methodology is applied across ten phases. The ML pipeline follows CRISP-DM applied to financial NLP: data understanding (CFPB complaint taxonomy analysis), preparation (text cleaning, financial entity normalisation, SMOTE class balancing), modelling (TF-IDF + Logistic Regression baseline, SVM, DistilBERT, FinBERT comparison), evaluation (weighted F1, per-class F1, confusion matrix, SHAP feature inspection, fairness analysis), and deployment integration. Software architecture follows domain-driven microservices decomposition (Dragoni et al., 2017) with JWT authentication, RBAC at the gateway, and OWASP API Security Top 10 compliance.

Key requirements:

Functional: multi-role authentication, full CRUD complaint lifecycle, real-time AI inference with SHAP explanations, chatbot-to-ticket escalation, FCA-timed automation triggers, interactive analytics. Non-functional: sub-500ms inference latency, JWT expiry within 60 minutes, UK GDPR-compliant data handling, FCA-compliant audit logging (complaint ID, timestamp, assigned agent, AI prediction, SHAP reference, override flag).

Scope and limitations:

Fully local deployment via Docker Compose. Microsoft Entra MFA simulated locally. Mobile app excluded from MVP. Platform targets UK retail banking and insurance only; investment and wholesale banking out of scope. All AI predictions require human agent review before action.

Skills and knowledge to acquire:

FinBERT fine-tuning on domain-specific complaint text; SHAP explainer selection per model type; FCA Consumer Duty compliance design; CFPB schema parsing and SMOTE class balancing; MongoDB aggregation pipelines; Recharts dashboard integration.

Datasets:

Primary: CFPB Consumer Complaint Database (Consumer Financial Protection Bureau, 2024) — available on Kaggle at kaggle.com/datasets/kaggle/us-consumer-finance-complaints. Contains 555,957 complaint records; approximately 66,000 rows retain usable narrative text after cleaning. Provides complaint category labels and a `timely_response` flag used to engineer priority labels. Public domain. Secondary: FinancialPhraseBank (Malo et al., 2014) — available on Kaggle at kaggle.com/datasets/sbhatti/financial-sentiment-analysis. Contains 4,846 expert-annotated financial sentences with positive, neutral, and negative sentiment labels. Used to train and evaluate the sentiment classification model. Licensed under CC BY-NC-SA. Both datasets are open source and require no ethics approval.

Technical architecture:

Next.js web app (5 roles) + chatbot + Recharts analytics dashboard. Nginx reverse proxy gateway. FastAPI backend services (auth, case management, AI inference, chatbot, automation, analytics). PostgreSQL + MongoDB hybrid storage. Fully local Docker Compose orchestration.

MVP vs Extension Scope

Must-build (MVP)	Extensions (time permitting only)
10 FastAPI microservices	Microsoft Entra MFA live integration
3 NLP models + SHAP explainability	RL-based complaint routing
Next.js web — 5 role portals	Generative/LLM chatbot upgrade
Hybrid chatbot — 6 categories	Advanced embedded analytics upgrade
5 automation rules	Redis caching layer
PostgreSQL + MongoDB storage	
Interactive analytics dashboard (Recharts)	
Docker Compose orchestration	
Full test suite + ML evaluation	

Risk Analysis

Risk	Severity (0-5)	Likelihood (0-5)	Action/Mitigation
Scope overrun — multiple services + web app + chatbot + automations in 19 weeks	4	4	MVP scope is fixed and reviewed at each phase gate. All extensions are explicitly optional. Depth

			and robustness beat feature count for top-band marking.
CFPB dataset class imbalance (debt collection dominates)	3	4	Apply SMOTE oversampling and class-weighted loss. Report per-class F1 scores explicitly. Class imbalance handling becomes part of ML methodology discussion.
FinBERT fine-tuning complexity	3	3	Use TF-IDF+LR as primary delivery model; FinBERT as comparison. Research value is in comparative evaluation, not solely in transformer accuracy.
Microservice integration failures	4	3	Health checks and service_healthy depends_on in Docker Compose. Services built and tested in isolation before integration.
Report delayed by development overrun	4	3	Literature review and methodology drafted by end of June alongside Phase 3. Report deadline (28 Aug) is 7 days before artefact deadline (4 Sep).

Consideration of Issues

Legal and Regulatory:

This project uses only public domain and CC-licensed datasets (CFPB, FinancialPhraseBank) and handles no real personal data. The system architecture is designed for UK GDPR compliance: data minimisation, purpose limitation, configurable audit log retention, JWT authentication, bcrypt password hashing, and RBAC. The SHAP explainability component directly addresses GDPR Article 22's right to explanation for automated decisions and FCA Consumer Duty PS22/9 algorithmic transparency requirements (FCA, 2023). FCA-compliant audit logs record complaint ID, timestamp, assigned agent, AI prediction, SHAP explanation reference, and human override flag.

Ethical considerations and AI bias:

Models trained on historical complaint data may inherit patterns that disadvantage certain customer groups. This project addresses this through per-class F1 analysis to detect performance disparities, SHAP inspection of top predictive features to identify protected-characteristic proxies, and explicit human override design. The system is a decision support tool, not an autonomous decision-maker, satisfying FCA human oversight requirements under Consumer Duty.

Diversity, inclusion, and accessibility:

Web application targets WCAG 2.1 AA guidelines. Chatbot supports plain language submission, reducing barriers for customers with lower digital literacy. Multi-channel design (web and chatbot interface) supports different user preferences and ability levels.

Professional:

Adheres to BCS Code of Conduct. Production deployment would require FCA notification, penetration testing, and independent audit before handling real customer data.

Ethics Approval Process

Will you be collecting data about identifiable individuals? NO

Will you be collecting sensitive personal data? NO

Will you be collecting data from vulnerable participants? NO

Will you be collecting data outside LTU campus or home address? NO

Will you be using datasets that are not open source? NO

Do you believe you need to make a full ethics application? NO

All datasets used (CFPB, FinancialPhraseBank) are publicly available open-source collections. No human participants. No personal data collected or processed.

Project Timeline

Phase	Dates	Key Activities	Deliverable
Phase 1 Design	Apr 21– May 8, 2026	Finalise architecture, define core service APIs, confirm 6-category financial complaint taxonomy, lock MVP scope, complete plan template	A1 Project Plan Due: 8 May 2026
Phase 2 Data & Schema	May 8– May 22, 2026	Download CFPB dataset, map to 6-class taxonomy, engineer priority labels from timely_response flag, download FinancialPhraseBank, design PostgreSQL schema (FCA audit fields), MongoDB collections, ERD	Clean datasets, DB schema, ERD
Phase 3 AI / ML	May 22– Jun 19, 2026	Text preprocessing, SMOTE, TF-IDF+LR baseline, SVM, DistilBERT/FinBERT, SHAP integration, full evaluation (weighted F1, per-class F1, confusion matrix, fairness analysis)	Trained models, evaluation report
Phase 4 Backend	Jun 19– Jul 17, 2026	Core FastAPI services (auth, case management, AI inference, analytics, automation, notification), Docker Compose orchestration, Postman collection, service-level unit and integration tests	Working APIs, integration tests
Phase 5 Web App	Jul 17– Jul 31, 2026	Next.js 5-portal app, agent dashboard with SHAP inline, chatbot widget, RBAC guards	Functional web application
Phase 6 Analytics	Jul 31– Aug 7, 2026	PostgreSQL reporting tables, interactive analytics dashboard in Next.js using Recharts (complaint volume, sentiment trend, FCA compliance rate, agent performance), metric validation	Analytics dashboard live in web app

Phase 7 Chatbot + Auto	Aug 7– Aug 14, 2026	Financial complaint intake flow (6 categories), NLP intent detection, escalation to ticket creation, 5 automation rules (fraud alert, FCA deadline reminder, negative-sentiment flag, resolution confirmation, daily manager summary)	Chatbot + automation engine
Phase 8 Testing	Aug 14– Aug 25, 2026	Unit, integration, security, performance tests; chatbot intent accuracy; ML evidence consolidation	Test suite + evidence
Phase 9 Report (parallel)	Jun 19– Aug 28, 2026	PARALLEL with Phases 4–8. Literature review + methodology drafted by end of June alongside Phase 4. Implementation section written during Phases 5–7. Results and discussion written Aug 14–28 once testing begins. Final proofread and submit.	A3 Report Due: 28 Aug 2026
Phase 10 Packaging	Aug 25– Sep 4, 2026	README with full run instructions, demo data setup, 10-minute demonstration video (voice-over or captions as required by brief), zip and submit	A2 Artefact Due: 4 Sep 2026

RACI Matrix

The RACI matrix below defines accountability for each major project activity. R = Responsible (does the work), A = Accountable (owns the outcome), C = Consulted (provides input), I = Informed (kept updated).

Activity	Student	Supervisor	Mod. Leader	Assessor	Notes
Project plan and proposal	R/A	C	I	I	Student owns and delivers
Dataset selection and preparation	R/A	C	I	I	CFPB + FinancialPhraseBank
ML model training and evaluation	R/A	C	I	I	NLP models + SHAP
Backend and system integration	R/A	C	I	I	FastAPI + Docker Compose
Web application development	R/A	C	I	I	Next.js + Recharts
Supervision and academic guidance	C	R/A	I	I	Dr Aliyu Lawal Aliyu
Testing and quality assurance	R/A	C	I	I	pytest, Jest, Postman
Assessment and marking	I	I	I	R/A	COM7016 marking team

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This assignment used generative AI in the following ways: brainstorming, research, planning, editing.